11 EXAM PRACTICE

1 An experiment is set up to measure the rate of hydrolysis of methyl ethanoate, CH_3COOCH_3.

$$CH_3COOCH_3 + H_2O \rightleftharpoons CH_3COOH + CH_3OH$$

The hydrolysis is very slow in neutral aqueous solution. When dilute hydrochloric acid is added, the reaction is faster.

What is the function of the hydrochloric acid?

A to increase the reaction rate by acting as a catalyst

B to make sure that the reaction reaches equilibrium

C to maintain a constant pH during the reaction

D to dissolve the methyl ethanoate [1]

(Total for Question 1 = 1 mark)

2 For the gaseous reaction $2X(g) + Y(g) \rightarrow Z(g)$, the rate equation is

$$\text{rate} = k[X]^2[Y]^0$$

If the pressure in the reaction vessel is doubled but the temperature remains constant, by what factor does the rate of reaction increase?

A 2 **B** 3 **C** 4 **D** 8 [1]

(Total for Question 2 = 1 mark)

3 The alkaline hydrolysis of RBr, where RBr = $(CH_3)_3Br$, takes place in two steps:

$RBr \rightarrow R^+ + Br^-$ SLOW

$R^+ + OH^- \rightarrow ROH$ FAST

Which of the following rate equations is consistent with this scheme?

A rate = $k[OH^-]$

B rate = $k[RBr]$

C rate = $k[RBr][OH^-]$

D rate = $k[R^+][OH^-]$ [1]

(Total for Question 3 = 1 mark)

4 The table gives data for the reaction between X and Y at constant temperature.

Experiment	[X] / mol dm^{-3}	[Y] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.3	0.2	4.0×10^{-4}
2	0.6	0.4	1.6×10^{-3}
3	0.6	0.8	6.4×10^{-3}

What is the rate equation for the reaction?

A rate = $k[X][Y]^2$

B rate = $k[X]^2[Y]$

C rate = $k[X]^2$

D rate = $k[Y]^2$ [1]

(Total for Question 4 = 1 mark)

5 The reaction of acidified aqueous potassium iodide with aqueous hydrogen peroxide

$$2I^-(aq) + H_2O_2(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

is thought to involve three steps:

$H_2O_2 + I^- \rightarrow H_2O + OI^-$ SLOW

$OI^- + H^+ \rightarrow HOI$ FAST

$HOI + H^+ + I^- \rightarrow I_2 + H_2O$ FAST

Which of the following conclusions **cannot** be deduced from this information?

A The acid is a catalyst.

B The reaction is first order with respect to the iodide ion.

C The rate determining step is $H_2O_2 + I^- \rightarrow H_2O + OI^-$

D The rate equation for the reaction is: rate = $k[H_2O_2][I^-]$ [1]

(Total for Question 5 = 1 mark)

6 Propanone and iodine react in aqueous acidic solution according to the following overall equation:

$$CH_3COCH_3 + I_2 + H^+ \rightarrow CH_3COCH_2I + 2H^+ + I^-$$

The experimentally determined rate equation for this reaction is:

$$\text{rate} = k[CH_3COCH_3][H^+]$$

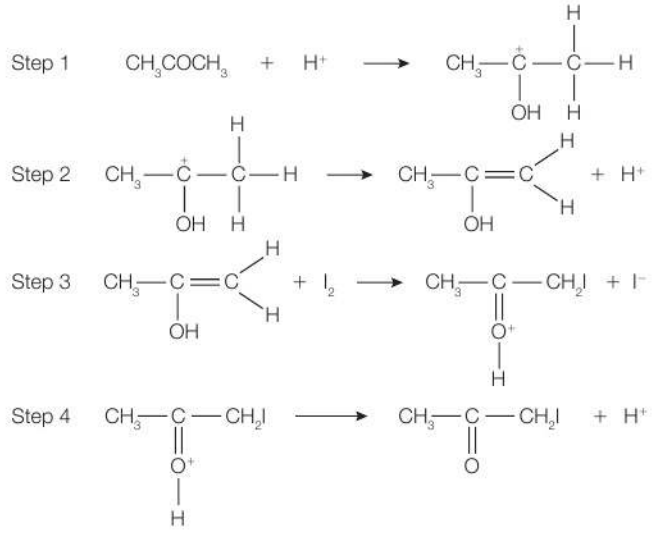
(a) With initial concentrations as shown, the initial rate of reaction was 1.43×10^{-6} mol dm^{-3} s^{-1}.

	Initial concentration/mol dm^{-3}
CH_3COCH_3	0.400
H^+	0.200
I_2	4.00×10^{-4}

Calculate a value for the rate constant, k, for the reaction. [3]

(b) Explain the effect on the rate of reaction of doubling the concentration of iodine, but keeping the concentrations of propanone and hydrogen ions constant. [2]

(c) The proposed mechanism for the overall reaction is:



Explain which of the four steps could be the rate-determing step. [3]

(Total for Question 6 = 8 marks)

7 Most chemical reactions involve two or more steps. The experimentally determined rate equation indicates which species are involved either before or in the rate-determining step.

(a) State what is meant by the term **rate-determining step**. [1]

(b) Hydrogen reacts with iodine monochloride in a two-step mechanism according to the following overall equation:

$H_2(g) + 2ICl(g) \rightarrow 2HCl(g) + I_2(g)$

The experimentally determined rate equation for this reaction is:

rate = $k[H_2(g)][ICl(g)]$

The rate-determining step is the first step in the mechanism for the reaction.

(i) Write an equation for the rate-determining step. [1]

(ii) Write an equation for the second step. [1]

(c) A series of experiments were carried out on the reaction A + B + C → products. The results are shown in the table.

Experiment	Initial concentrations/ mol dm^{-3}			Initial rate/ mol dm^{-3} s^{-1}
	[A]	[B]	[C]	
1	0.100	0.100	0.100	6.20×10^{-4}
2	0.100	0.200	0.100	6.20×10^{-4}
3	0.100	0.100	0.200	2.48×10^{-3}
4	0.200	0.100	0.100	1.24×10^{-3}

(i) Determine the order of reaction with respect to A, B and C. Show how you obtain your answers. [3]

(ii) Write the rate equation for the reaction. [1]

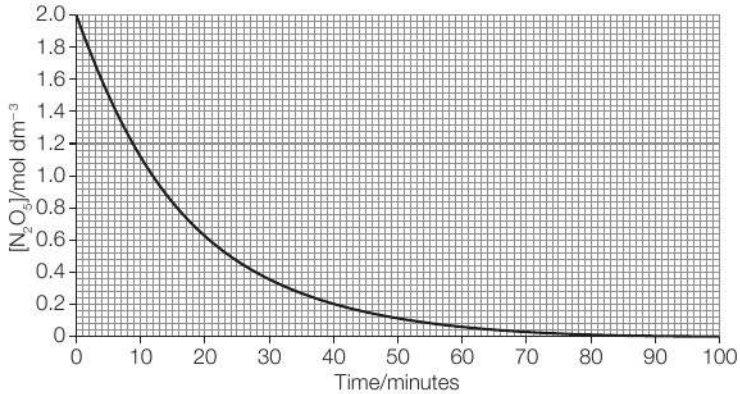
(iii) Which reactant is unlikely to be in the rate determining step? [1]

(Total for Question 7 = 8 marks)

8 Nitrogen pentoxide decomposes on heating to form nitrogen tetroxide and oxygen. The equation for the reaction is:

$N_2O_5(g) \rightarrow N_2O_4(g) + \frac{1}{2}O_2(g)$

The progress of the reaction can be followed by measuring the concentration of N_2O_5 present. The graph shows the results obtained in an experiment conducted at constant temperature.



(a) (i) Use the graph to predict the rate of reaction after 20 minutes and after 90 minutes. [3]

(ii) Deduce the rate of production of oxygen after 20 minutes? [1]

(b) (i) Plot on the graph two successive half-lives for this reaction. [2]

(ii) Deduce the order of reaction with respect to N_2O_5. Justify your answer. [2]

(iii) Write a rate equation for the reaction. [1]

(c) (i) Calculate a value for the rate constant, k, for this reaction. [3]

(ii) Deduce the initial rate of reaction when the initial concentration of N_2O_5 is 1.50 mol dm^{-3}. Assume the reaction is carried out under the same conditions of temperature and pressure. [2]

(Total for Question 8 = 14 marks)

9 Nitrogen dioxide, NO_2, is decomposed on heating into nitrogen monoxide, NO, and oxygen. The proposed mechanism for this reaction is:

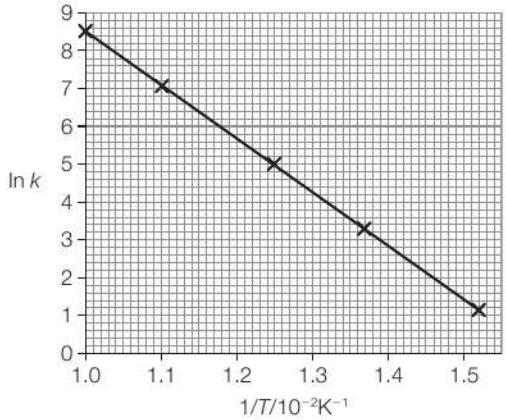
$NO_2 + NO_2 \rightarrow NO_3 + NO$ SLOW

$NO_3 \rightarrow NO + O_2$ FAST

(a) (i) Write an overall equation for the decomposition of nitrogen dioxide. [1]

(ii) Write a rate equation for the reaction. [1]

(b) The rate constant, k, for the decomposition was determined at several different temperatures. The results obtained were used to plot the graph shown below.



(i) Use the graph and the expression

$$\ln k = -\frac{E_a}{R}\frac{1}{T} + \ln A$$

to calculate the activation energy, E_a, for the thermal decomposition of nitrogen dioxide. Give your answer to an appropriate number of significant figures. [4]

(ii) A vessel of volume 2.00 dm^3 is filled with 4.00 mol of ntitrogen dioxide at a temperature of 650 K. The initial rate of reaction was found to be 12.64 mol dm^{-3} s^{-1}.

Calculate a value for the rate constant, k, at this temperature. [3]

(Total for Question 9 = 9 marks)